

Métodos Matemáticos de la Física II

Partial Differential Equations

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(Dated: June 3, 2026)

I. CLASSIFICATION

Consider Partial Differential Equations (PDEs) defined by

$$a^{ij}(x, u(x), (\partial u)(x))(\partial_i \partial_j u)(x) + f(x, u(x), (\partial u)(x))$$

for all $x = (x^1, \dots, x^n) \in D \subset \mathbb{R}^n$ with solutions $u : D \ni x \rightarrow u(x) \in \mathbb{R}$. Here $(\partial u)(x) = ((\partial_1 u)(x), \dots, (\partial_n u)(x)) \in \mathbb{R}^n$. Also, $f : D \times \mathbb{R} \times \mathbb{R}^n \ni (x, u(x), (\partial u)(x)) \rightarrow f(x, u(x), (\partial u)(x)) \in \mathbb{R}$.

For each $x \in D$ and solution u , the matrix of coefficients

$$a^{ij}(x, u(x), (\partial u)(x)) \in \mathbb{R}$$

is assumed to be symmetric. We will say that the PDE is:

- *elliptic* at $(x, u(x), (\partial u)(x))$ if the eigenvalues of the mentioned matrix are non-zero and of the same sign,
- *hyperbolic* at $(x, u(x), (\partial u)(x))$ if the eigenvalues of the mentioned matrix are non-zero but of different signs, and
- *parabolic* at $(x, u(x), (\partial u)(x))$ if at least one of the eigenvalues of the mentioned matrix is zero and the non-zero ones have the same sign.

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